

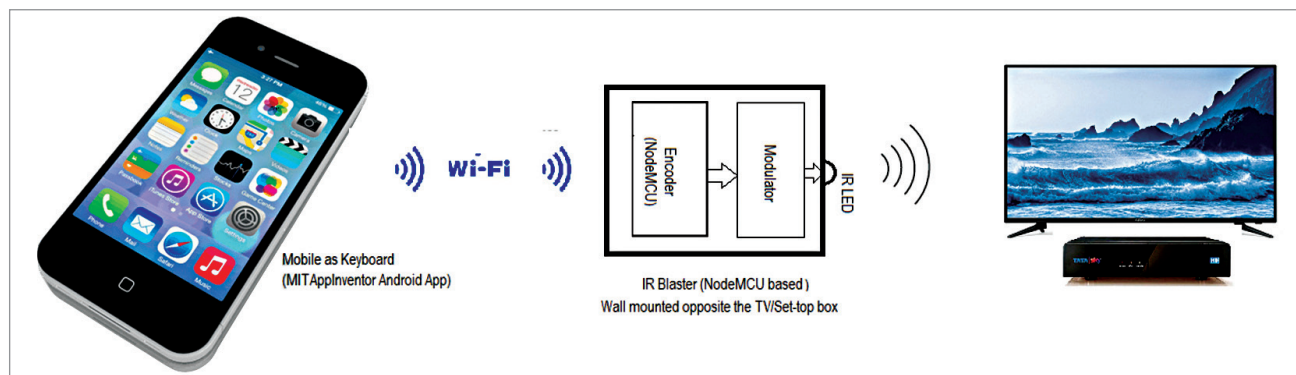


Fig. 3: Functional diagram of Wi-Fi remote



Fig. 4: Android app icon

Downloads folder to begin the installation.

3. Change the 'Allow installation of apk files from unknown sources' setting back to the original state to restore the original settings.

The app should now be ready for use. The app icon will look as shown in Fig. 4.

The app - first look

The app begins by displaying Opening Screen message. You are presented with two options: Settings (by way of a sprocket icon) and Select TV button. The app is designed to work on multiple TVs you may have at home. The IR transmitter module is separate for each TV and has its own IP address.

Press the Settings option (sprocket icon) in the app to open the Settings screen. Here you can set up the TV names and IP address of respective IR-Blasters (explained later in the article). The app supports multiple TVs/IR-Blasters you may have. The Settings screen allows you to Add, Modify, Delete, and Clear All.

You can return to the opening screen by pressing the Cancel/Back

button. The list of available TVs is shown, and you can select a TV name to modify its existing TV IP address.

The option Select TV will display the names of the TVs set up in your home. The desired TV should be selected so that the mobile is linked to the respective IR-Blaster. Now the function selections made from the mobile phone will be received by the IR-Blaster for further transmission over infrared beams to the TV.

Once the TV name is selected, the screen will change to Channel Selector screen. The name of the TV will be displayed in the title bar. On pressing the power button, the set-top box and TV will receive IR signals for switching on or off the two gadgets simultaneously. As such, there is no need to handle two separate remotes.

The TV will also switch to your favourite channel (set in the IR-Blaster code) on start-up instead of the ad screen that is normally shown when TataSky starts. The volume control buttons for controlling the set-top box and the TV are available separately. A mute button is also available for shutting off the sound, when required. The number keys can be used for typing any channel number while the Channel Up and Channel Dn buttons will change the channel to the next and previous channel, respectively.

To make life easier, while using this Wi-Fi remote, there is no need to remember the channel numbers. Each TV channel has been assigned a separate icon, and with one press of the relative icon the set-top box will change over to the desired channel. To organise the icons, the channels are grouped under their respective genre. On pressing the Select Genre button,

the screen will change to Select Genre screen, which presents a selectable list of the various group types.

On selecting a genre, the screen will change and present up to 25 icons of the available channels in that group. The Volume, Mute, Power, and Select Genre buttons will always be available. On selecting any icon, the IR-Blaster will transmit signals for changing over to the selected channel. The channel number for each station has to be entered in the app.

You can go back to the Channel Selector screen by choosing the Channel Selector option from the Select Genre list. Another TV can be selected by selecting the Select TV button on the Channel Selector screen. If you want to control the TV in your kid's room, well, now you can do it provided the same Wi-Fi network is available across the house.

Building the app

The app has been built using the MIT App Inventor 2, which is an internet based Android app development tool. The interface is totally graphical including the code. The tool is free and can be accessed by anyone having a Gmail account.

The complete source code for the app is available for download from the EFY website as an AIA file. This file can be viewed in MIT App Inventor 2.

After downloading the file to your computer, start MIT App Inventor and import the file using the Import Project option in the Projects top-level menu. Go through various screens and blocks of the given project (AIA file) to get a feel of the structure of the program. This will help you in customising the app to your needs. (More details on